

SAFETY DATA SHEET

Formula 204

SECTION 1. PRODUCT AND COMPANY IDENTIFICATION

Sold By: Water Sciences – 3208 Caprice Court, Fort Wayne, IN 46808 – (260) 485-4655

FOR 24 HOUR EMERGENCY RESPONSE CALL CHEMTREC (800) 424-9300

Product Number: 204
Product Name: Water Sciences Formula 204
Synonym(s): Water Treatments Compound, Liquid
Recommended Use: Boiler Feed Water Oxygen Scavenger



SECTION 2. HAZARDOUS IDENTIFICATION

GHS CLASSIFICATION

Classification

Acute Toxicity, Oral	Category 4
Acute Toxicity, Dermal	Category 4
Skin Corrosion/Irritation	Category 2
Serious Eye Damage/Eye Irritation	Category 2A
Acute Toxicity, Inhalation	Category 4
Hazardous to the Aquatic Environment, Acute Hazard	Category 3

Signal Word: **WARNING!**

GHS Label:



Hazard Statements

H302	Harmful if swallowed.
H312	Harmful in contact with skin.
H315	Causes skin irritation.
H319	Causes serious eye irritation.
H332	Harmful in inhaled.
H402	Harmful to aquatic life.

Precautionary Statements

P260	Avoid breathing dust/fume/gas/mist/vapors/spray.
P264	Wash thoroughly after handling.
P270	Do not eat, drink, or smoke when using this product.
P273	Avoid release to the environment.
P280	Wear protective gloves/protective clothing/eye protection/face protection.
P301+312	IF SWALLOWED: Call a Poison Center or doctor/physician if you feel unwell.
P302+352	IF ON SKIN: Wash with plenty of water.
P304+340	IF INHALED: Remove victim to fresh air and keep comfortable for breathing.
P305+351+338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contacts if present and easy to do. Continue rinsing.
P312	Call a Poison Center or a doctor/physician if you feel unwell.
P321	Specific treatment (see first aid on this label).
P330	Rinse mouth.
P332+313	If skin irritation occurs, get medical advice/attention.
P337+313	Take off immediately all contaminated clothing and wash it before reuse.
P501	Dispose of contents/container to a licensed waste hauler.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

The precise composition of this product is proprietary information. In the even of a medical emergency, a complete disclosure will be provided to medical personnel.

Component Name	CAS #	% Wt	OSHA PEL	ACGIH TLV
Sodium Sulfite	7757-83-7	8-15	Not Established	Not Established
Sodium Bisulfite	007631-90-5	12-20	Not Established	5 mg/m ³
Water	7732-18-5	Balance	Not Established	Not Established

SECTION 4. FIRST AID MEASURES

Exposure Route	
Eye Contact	Eye irritation. Immediately flush eyes with plenty of water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Call a physician or poison control center immediately.
Skin Contact	Itching or burning of the skin. Take off immediately all contaminated clothing. Wash off IMMEDIATELY with plenty of water for at least 15-20 minutes. Get medical attention immediately! Wash clothing separately before reuse. Destroy or thoroughly clean contaminated shoes.
Inhalation	Nasal irritation, headache, dizziness, breathing difficulty, cyanosis, tremors, weakness, red flushing of face, irritability.
Ingestion	Call a physician or poison control center immediately. Do not induce vomiting. Immediately rinse mouth and drink plenty of water. If vomiting occurs, keep head low so that stomach content doesn't get into the lungs. Never give anything by mouth to an unconscious person. Do not use mouth-to-mouth method if victim ingested the substance.

SECTION 5. FIREFIGHTING MEASURES

Suitable Extinguishing Media	Use extinguishing measures that are appropriate to surrounding fire conditions.
Special Firefighting Procedures	Do not flush down sewers or other drainage systems. Fire fighters should enter the area only if they are protected from all contact with the material. Full protective clothing, including self-contained breathing apparatus, coat, pants, gloves, boots and bands around legs, arms, and waist, should be worn. No skin surface should be exposed.
Unusual Firefighting Hazards	Combustion Products: Thermal decomposition products may include oxides of potassium and sulfur.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Personal Precautions	Keep unnecessary people away; isolate hazard area and deny entry. Stay upwind; keep out of low areas. (Also see Section 8).
Environmental Precautions	Contain spill if it can be done with minimal risk. Prevent liquid from entering drains, sewers or waterways. Notify proper authorities.
Methods for Cleanup	Small spills: Take up with sand or other noncombustible absorbent material and place into containers for later disposal. For large spills, dike far ahead of contaminated runoff for later disposal.

SECTION 7. HANDLING AND STORAGE

Handling	Do not get in eyes, on skin or on clothing. Do not breathe vapors or mists. Keep container closed. Use only with adequate ventilation. Use good personal hygiene practices. Wash hands before eating, drinking, or smoking. Remove contaminated clothing and clean before reuse.
Storage	Store in tightly closed containers in cool, dry, well-ventilated area away from heat, sources of ignition and incompatibles. Store at ambient or lower temperature. Store out of direct sunlight. Keep containers tightly closed and upright when not in use. Protect against physical damage. Empty containers may contain harmful residue or vapors.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

EXPOSURE LIMITS

Component	CAS #	OSHA (PEL)	ACGIH (TLV)
Sodium Sulfite	7757-83-7	Not Established	Not Established
Sodium Bisulfite	7631-90-5	Not Established	5 mg/m ³
Water	7732-18-5	Not Established	Not Established

Engineering Controls	Provide general or local exhaust ventilation systems to maintain airborne concentrations below OSHA limits (Sec. 2). Local exhaust ventilation is preferred because it prevents contaminant dispersion into the work area by controlling it at the source.
Monitoring	Follow OSHA respirator regulations (29 CFR 1910.134) and, if necessary, wear a MSHA/NIOSH-approved respirator. Select respirator based on its suitability to provide adequate worker protection for given working conditions, level of airborne contamination, and presence of sufficient oxygen. For emergency or non-routine operations (cleaning spills, reactor vessels, or storage tanks), wear a SCBA. Warning! Air-purifying respirators do not protect workers in oxygen-deficient atmospheres.
PERSONAL PROTECTIVE EQUIPMENT (PPE)	
Eye Protection	Safety glasses with side-shields conforming to EN166 Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU).

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION - continued

PERSONAL PROTECTIVE EQUIPMENT (PPE) - continued	
Skin Protection	Avoid skin contact. Wear gloves impervious to conditions of use. Additional protection may be necessary to prevent skin contact including use of apron, face shield, boots or full body protection. A safety shower should be located in the work area.
Respiratory Protection	If exposure limits are exceeded, NIOSH approved respiratory protection should be worn. A NIOSH approved respirator for organic vapors is generally acceptable for concentrations up to 10 times the PEL. For higher concentrations, unknown concentrations and for oxygen deficient atmospheres, use a NIOSH approved air-supplied respirator. Engineering controls are the preferred means for controlling chemical exposures. Respiratory protection may be needed for non-routine or emergency situations. Respiratory protection must be provided in accordance with OSHA 29 CFR 1910.134.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	Clear light pink liquid	Solubility	Disperses
Odor	Mild odor	Autoignition Temperature	Not known
Odor Threshold	Not known	Decomposition Temperature	Not known
pH @ 25° C	6.0 – 7.0	VOC Content	% Volatile: ~ 65% (water)
Melting Point/ Freezing Point	Not known	Boiling Range	100° C
Flashpoint	> 200° F	Flammable Limits, Upper & Lower	N/A
Specific Gravity	1.25 g/ml @ 20° C	Vapor Pressure	Not known
Vapor Density (Air = 1)	Not known	Viscosity	20-40 cP @ 25° C

SECTION 10. STABILITY AND REACTIVITY

Stability	Stable under normal conditions of storage and handling.
Conditions to Avoid	Contact with incompatible materials.
Hazardous Decomposition/Byproducts	Thermal decomposition products may include oxides of potassium and sulfur.
Hazardous Polymerization	Will not occur.
Polymerization Conditions to Avoid	None
Incompatibilities	Strong acids and oxidizers

SECTION 11. TOXICOLOGICAL INFORMATION

Likely Route of Exposure	Inhalation, Skin, Eye, Ingestion
Inhalation	May cause severe irritation with possible lung damage (pulmonary edema).
Eye Contact	May cause severe conjunctival irritation and corneal damage.
Skin Contact	May cause reddening, blistering, or burns with permanent damage. Harmful if absorbed through the skin. May cause allergic skin reaction.
Ingestion	May cause gastrointestinal irritation with nausea, vomiting, and diarrhea.
Acute Toxicity Value	See Health Hazards Below. Acute Toxicity Values: - Oral LD50 (Rat) = > 2000 mg/kg - Dermal LD50 (Rabbit) = > 2000 mg/kg - Inhalation LC50 (Rat) = 800 ppm/4 hrs
Chronic (Long-Term) Effects	Prolonged or repeated exposure may cause dermatitis, and sensitization reactions. Exposure to asthmatic, atopic and sulfite sensitive individuals may result in severe bronchoconstriction and reduced levels in forced expiratory volume. Decomposition of sodium bisulfite solutions may release toxic and hazardous fumes of sulfur oxides, including sulfur dioxide, which may cause permanent pulmonary impairments from acute and chronic exposure. The Immediately Dangerous to Life or Health (IDLH) level for SO2 is 100 ppm.

TOXICITY:

Component Name	LD50	LC50
Sodium Sulfite	Oral (Rat): 2610 mg/kg	Inhalation (Rat): > 5.5 mg/l 4hrs
Sodium Bisulfite	Not established	Not established
Water	Not established	Not established

Reproductive Effects	N/A	Embryotoxicity	N/A
Teratogenicity	N/A	Sensitization to Product	N/A
Mutagenicity	N/A	Synergistic Products	N/A
Carcinogenicity	N/A		

SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity	Aquatic Toxicity (e.g., 96 Hr. TLM): - Bluegill: 240 mg/l, LC50, 48 hours - Fathead Minnow: 85 mg/l, LC50 96 hours - Ceriodaphnia Magna: 105 mg/l, LC50, 48 hours
Mobility	Slight
Degradability	Biodegradation: Biodegradable
Bioaccumulation	Information not available

SECTION 13. WASTE DISPOSAL CONSIDERATIONS

Dispose of all material in accordance with Federal, State, and Local laws. This material does not meet RCRA's characteristic definition of corrosivity.

SECTION 14. TRANSPORT INFORMATION

Proper Shipping Name: Not DOT Regulated
Other Shipping Info: Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code.

SECTION 15. REGULATORY INFORMATION

EPA REGULATIONS					
TSCA Status: The components of this product are listed on the TSCA Inventory					
SARA TITLE III SECTION 302/304 EXTREMELEY HAZARDOUS SUBSTANCE: No chemicals in this material are subject to the reporting requirements.					
SARA TITLE III SECTION 311/312 HAZARD CATEGORIZATION:					
	Acute	Chronic	Fire	Pressure	Reactive
	X	X	N/A	N/A	N/A
SARA TITLE III SECTION 313 SUPPLIER INFORMATION: No chemicals in this material are subject to the reporting requirements.					
CALIFORNIA PROPOSITION 65: No chemicals in this material are subject to the reporting requirements.					

CERCLA SECTION 102(a) HAZARDOUS SUBSTANCE:

Component Name	CAS #	% by Wt.	RQ (Lbs.)
Sodium Bisulfite	7631-90-5	15-20	5000

SECTION 16. OTHER INFORMATION

HMIS Health	1
HMIS Flammability	0
HMIS Reactivity	0

SDS Prepared/Reviewed by M. Fuze

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Employers should use this information only as a supplement to other information gathered by them and should make independent judgment of suitability of this information to ensure proper use and protect the health and safety of employees. This information is furnished without warranty and any use of the product not in conformance with this Safety Data Sheet, or in combination with any other product or process, is the responsibility of the user.

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